Calorie Expenditure: Intermediate Status Report

Project overview

At this checkpoint, the full pipeline is built and validated: EDA, cleaning decisions, reprocessing, 5-fold CV, baselines and a gradient-boosting ensemble with a 5-fold CV RMSLE of 0.0593, plus a Kaggle submission.

Item	Value
Competition	Predict Calorie Expenditure (Kaggle, S5E5)
Task	Regression: predict calories burned during a workout
Target	Calories (kcal), continuous, 1-314
Metric	RMSLE, lower is better
Repository	https://github.com/mangayankuba-dotcom/Predict-Calories-Expenditure-

Member	Role	Notebook sections / commits
Yankuba Manga	Data and EDA lead	Data loading, description, EDA, missing values and anomalies
Syed Ahsan Iqbal	Features and modelling	Preprocessing, feature engineering, baselines, LightGBM / XGBoost / CatBoost CV
Brefo Emmanuella Serwuaah	Assembling, evaluation and submission	Ensemble, error analysis, Kaggle submission, report and slides

What has been done

Data. Train has 750,000 rows × 9 columns; test has 250,000 rows × 8 (no target). There are 7 features: Sex (binary categorical: male / female) and six numeric ones: Age (years), Height (cm), Weight (kg), Duration (workout minutes, 1-30), Heart Rate (average bpm during the workout) and Body Temp (°C during the workout). id is a row identifier and is not used. The data is synthetic, generated by Kaggle from the public *Calories Burnt Prediction* datasets.

Key EDA findings

- The target is right-skewed (mean 88, median 77, skew 0.54), with about 3,000 rows at ≤ 2 kcal. Because the metric is RMSLE, all models are trained on log1p(Calories) with an RMSE loss, which optimizes the metric directly.
- Duration (r = 0.96), Heart_Rate (0.91) and Body_Temp (0.83) dominate. The relation is multiplicative (calories ≈ intensity × time) and its shape differs by sex.
- Height and Weight alone are almost uncorrelated with the target but matter together with sex and heart rate. Age has a weak positive effect (0.15).
- Sex is balanced (50.1% female). Duration is roughly uniform, Body Temp is strongly left-skewed.
- Train and test distributions are practically identical, so a random K-fold split represents the test set well.

Missing values and anomalies. There are no missing values in train or test. Everything below is kept, because none of it is a data error, and no flag features are added.

Issue	Count	Decision	Reason
Exact duplicate rows	2,841 (0.4%)	Keep	Legitimate draws from the same generator as the test set
Same features, different target	4,862 combinations	Keep	Irreducible noise; no model can remove it
Body_Temp IQR outliers (37-38 °C)	14,919 (2%)	Keep	Physiologically valid for short workouts; test has the same tail
Height, Weight, Heart_Rate, Calories outliers	< 150 each	Keep	Plausible human ranges (BMI 12-46)
Very low targets (≤ 2 kcal)	~3,000	Keep	Short workouts; they weigh heavily in RMSLE

Predictions are clipped to the training target range before submission.

Preprocessing. Sex is encoded as male = 1, female = 0; one-hot encoding would add a redundant column. Tree models are not scaled. The Ridge baseline uses StandardScaler inside a pipeline, fitted on training folds only to avoid leakage. We added 18 physiology-based features (25 in total): BMI, body surface area, products such as Duration × Heart_Rate, heart-rate reserve, temperature rise, sex interactions and the Keytel et al. (2005) energy-expenditure formula (r = 0.92 with the target).

Validation. We use 5-fold KFold (shuffled, seed 42) on the full training set, with the same folds for every model. RMSLE is reported per fold and on all out-of-fold predictions. A random split

fits because rows are independent, there is no time or group structure, and train and test share one distribution.

Models. Baselines: a constant (the mean of log1p(Calories)) and Ridge regression on raw and on engineered features. Main models: LightGBM, XGBoost and CatBoost with early stopping. Ensemble: non-negative least-squares weights fitted on out-of-fold predictions.

Current scores

The ensemble reaches a 5-fold CV RMSLE of 0.0593, 93.8% below the constant baseline. Engineered features give the largest single gain: Ridge drops from 0.1799 to 0.0649.

Model	CV RMSLE (5-fold OOF)	Std across folds	vs constant
Constant baseline (mean of log1p)	0.96323	0.00102	—
Ridge, 7 raw features (scaled)	0.17991	0.00072	−81.3%
Ridge, 25 engineered features (scaled)	0.06489	0.00048	−93.3%
LightGBM	0.06016	0.00040	−93.8%
XGBoost	0.05979	0.00039	−93.8%
CatBoost	0.05960	0.00031	−93.8%
Ensemble (weights 0.17 / 0.33 / 0.50)	**0.05930**	—	−93.8%

Kaggle: Public [score to be added] · Private [score to be added]. Submission: the NNLS ensemble of LightGBM, XGBoost and CatBoost.

Error analysis shows most of the remaining error comes from 1–3 minute workouts, where a 1 kcal mistake is a large relative error.